Walking Gigabytes

With a slew of funky and ultra portable storage devices, you now have the freedom you need to take your data with you, no matter where you go or what your applications

As the lines between our professional and personal spaces blur, we feel a greater need to carry our data from one sphere to another. So how do you transfer data that's more than a gigabyte? The answer lies in external portable devices. These devices are becoming smaller and at the same time their capacity to store data is increasing at a rapid rate. Portable devices such as hard drives, Zip drives, thumb drives and CD-Writers are also ideally suited to store your important data since they can reliably hold greater amounts as compared to the floppy disk. Add to that the fact that they are compact, light in weight and support a plethora of interface standards such as USB 2.0, FireWire, SCSI and so on. Even in a country like ours, where the floppy disk still rules, people are slowly shifting to portable solutions. So say hello to these new sleek members of the storage family—the external portable devices.

Digit Test Process

We tested all the portable devices on our test machine comprising a Pentium 4 2.8 GHz processor on an Intel D850EMV2 motherboard with 512 MB of RDRAM. The graphics card used was a Visiontek GeForce3 with 64 MB of DDR memory and a Seagate ATA IV. A 40 GB hard disk was installed on the primary IDE channel. All the test drives were connected to the appropriate ports (USB 1.1 or 2.0) and the necessary drivers and benchmarking utilities were loaded on to the drive after installing Windows XP. The latest drivers for the mainboard and the IDE channel were installed for maximum performance gain.

Test methodology

The drives were tested on four primary parameters: features (30 per cent), performance (40 per cent), value for money (20 per cent) and warranty and support (10 per cent). The weightages allotted to each parameter was used to calculate the overall scores.

Features: Being portable, the device should weigh as less as possible, support all major operating systems and should not be bulky. Apart from this, the build quality also matters. We took all these factors into account as we monitored the devices through our feature analysis.

Performance: We used the SiSoft Sandra 2002 Pro benchmark. It gave vital information about the drives such as the data transfer rates, the sequential and random read as well as write speeds, and the average access time. SiSoft Sandra 2002 continuously reads and writes data on to the disk and at the end, gives the separate as well as the overall scores.

To test the time a drive takes to read data, we transferred a set of assorted files to the tune of 1.27 GB from the system hard drive to the device under test. The assorted write time was then noted as the time taken to copy the same files back to the system drive. The sequential tests (both read and write) of the drives consisted of transferring a single large file between the portable and the fixed hard drive installed in the computer system.

Warranty and support: Warranty and support is very important for any device, especially for a device meant to be used on the move. Here we took into account the number of years of replacement warranty provided by the manufacturer.

Value for money: The value for money quotient was calculated by determining the sum of performance and features divided by the price of the device.

How they fared

In our test analysis, we looked into different features of the device such as weight, dimensions, storage capacity, interface and so on. The documentation provided with each of the drives was also checked. For evaluating performance, we conducted synthetic as well as real world tests and the performance figures were calculated accordingly.

Performance

An external storage device should not only store data but must also transfer the same at a fast rate. To gauge the transfer rates and evaluate the drives on a technical level, we ran synthetic as well as real world tests. SiSoft Sandra 2002 Pro served as our benchmark, while we transferred a set of files from the system drive to the device under test to paint a real-world picture.

SiSoft Sandra 2002 Pro: Here we noted different scores such as the sequential and random read and write scores, the drive index and the average access time. These numbers give you a probable transfer rate that the drive is capable of delivering. The sequential read rate of a storage system, for example, tells you how fast a drive can read large single files, say the